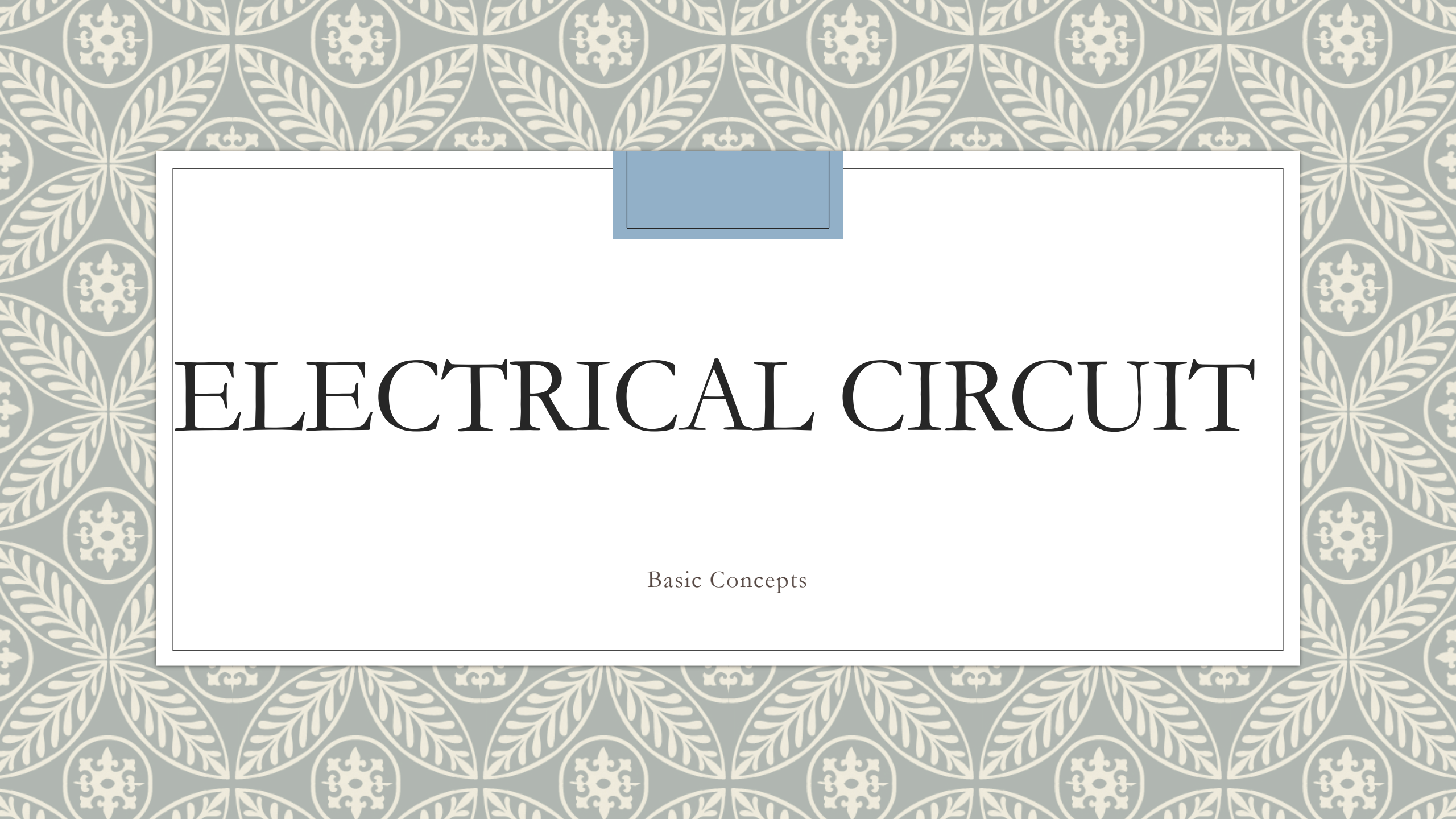


ELECTRICAL CIRCUIT

Basic Concepts

Historical Profiles

Alessandro Antonio Volta (1745-1827)

- Italian Physicist, Noble family in Como
- Invented Electric Battery (1796) and Capacitor
- Battery provides first continuous flow of electricity
- During 1800's, his work marked the beginning of electric Circuit theory.



Andre-Marie Ampere (1775-1836)

- French Mathematician and Physicist, Lyons.
- He laid the foundation of Electrodynamics
- Defined Electric Current and developed a way to measure it in the 1820s
- Mastered Latin in a few weeks at age 12 to learn math.
- Formulated the law of electromagnetics. He is a prolific writer.
- He invented electromagnet and the ammeter

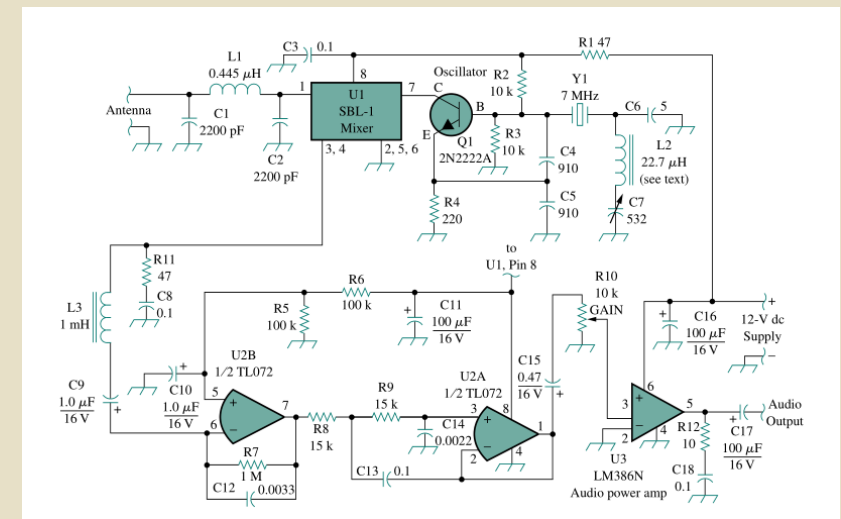
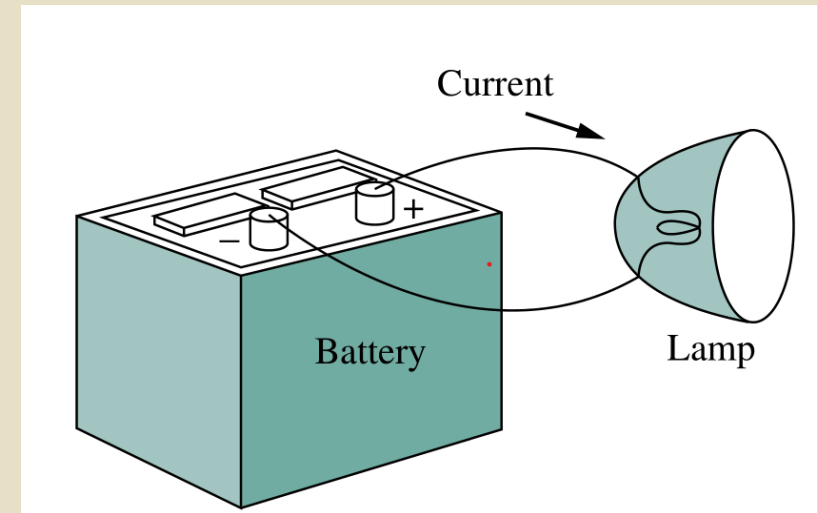


Electric Circuit- Why?

- All branches of Electrical Engineering are built upon two fundamental theories. Electric Circuit theory and Electromagnetic theory.
- Many branches of Electrical Engineering, such as, power, electric machines, control, electronics, communications and instrumentations are based of Electric circuit theory.
- Circuit theory is also valuable to students specializing in other branches of physical sciences, because circuits are a good model for the study of energy systems, which involves applied math, physics, and topology.
- To transfer Energy from one point to another, interconnection between devices is required. Consisting an electric circuit. Where each component of the circuit is known as an element.

Electric Circuit- What?

- An **Electric Circuit** is an interconnection of electrical elements.
- Simple electrical circuit is shown in the picture which consists of three basic components: a battery, a lamp, and connecting wires. Can be used as a torch light, a search light and so forth.
- A complicated circuit of a radio receiver is shown in the figure. Although it seems very difficult, but the basics are same and easy.
- We will analyze those circuits after knowing some basics- such as: charge, current, voltage, circuit elements, power, and energy.



International Systems of Units- SI (1960)

TABLE 1.1 The six basic SI units.

Quantity	Basic unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd

TABLE 1.2 The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^2	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

Electric Charge

- Charge is an electrical property of the atomic particles of which matter consists, measured in coulombs.(C)
- Doorknob shock, hair stick up using balloon experiment, lightening strike.- Static Electricity
- Static Electricity occurs when an object obtains a net amount of positive or negative electric charge, creating an imbalance that wants to be returned to equilibrium.
- Two piece of paper stuck to a table experiment. (First case, both stole negative charge)
- Atoms are electrically neutral as electron and proton are equal. Protons are fixed but some electrons are free to move around. Free electrons reside in a atom's outer shell as valence electrons and are easily plucked off and carried around, when acted upon by an outside force.
- Imbalance of electrical charge (free electrons) causes the free electrons to move at the first place.
- Negative charge- too many electrons, positive charge- missing electrons.

Electric Charge contd...

◦ Imbalance can be created in several ways:

1. **Charging by friction:** Rubbing a glass using a cloth, leaving the glass overall positive by taking negative charges. Fur with amber- that attracts hair and feathers- fur takes negative charge and makes amber positive. No charge is created, but overall charge is equal. Law of conservation of electric charge.
2. **Charging by contact:** if negatively charged glass is taken near to neutral glass rod, some charges is transferred to the neutral rod as long as they become equal.
3. **Charging by Induction:** If negatively charged object is bring near to a positively charged object, negative charge is accumulated to one side and other will be positively charged. If we break them, two different charged object is formed. By polarization
4. **Grounding:** If charged object is connected to a large object. Net distributed charge will be neutral.

Electric Charge Contd...

- Charge of a Electron: $-1.6 \times 10^{-19} \text{ C}$
- The number of electrons in one negative coulomb of charge: 6.24×10^{18}
- Charge of electrons is elementary charge and denoted by 'e', natural charges are integral multiply of 'e'
- $F = k \frac{q_1 q_2}{r^2}$ (Negative value attracts, and vice versa)
- k = Coulomb's constant= depend on the medium surrounding charges.= $9 \times 10^9 \text{ Nm}^2/\text{C}^2$
- Realistic valued/ Laboratory values of charges are on the order of pC, nC or μC

Electric Current

When a conducting wire (consisting of several atoms) is connected to a battery (a source of electromotive force), the charges are compelled to move; positive charges move in one direction while negative charges move in the opposite direction. This motion of charges creates electric current.

Electric current is the time rate of change of charge, measured in amperes (A).

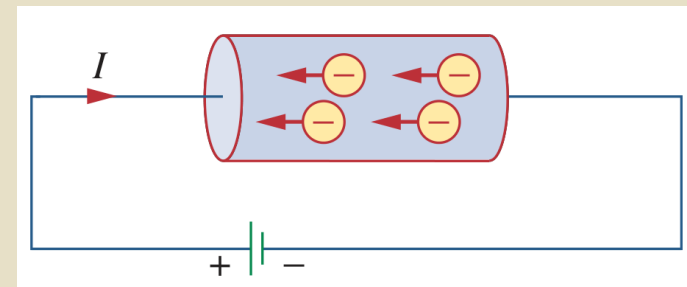
Conventionally we take the current flow as the movement of positive charges. That is, opposite to the flow of negative charges. It was introduced by American scientist Benjamin Franklin(1706–1790). Although we now know that current in metallic conductors is due to negatively charged electrons, we will follow convention.

Mathematically, the relationship between current i , charge q , and time t is: $i = \frac{dq}{dt}$ and $q = \int_t^{t_0} i dt$

where current is measured in amperes (A), and 1 ampere = 1 coulomb/second

Q = charge transfer between t_0 to t time

NB: A convention is a standard way of describing something so that others in the profession can understand what we mean. We will be using IEEE conventions throughout this book.



Electric Current Contd...

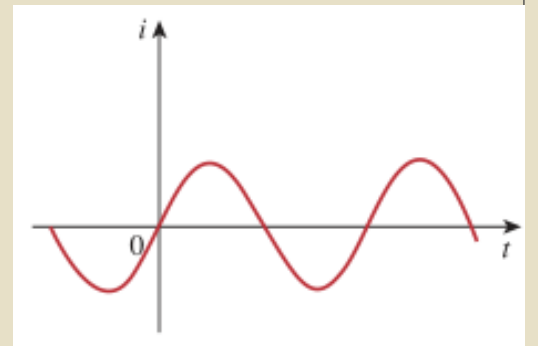
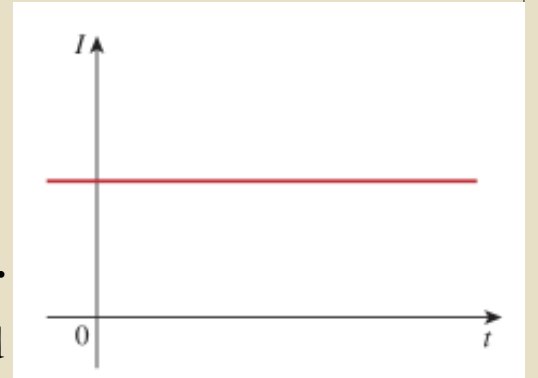
Direct Current: If the current does not change with time, but remains constant, we call it a direct current (dc)

A direct current (dc) is a current that remains constant with time.

Alternating Current: **An alternating current (ac) is a current that varies sinusoidally with time.**

Such current is used in your household to run the air conditioner, refrigerator, washing machine, and other electric appliances

Conventionally, Capital 'I' represents DC current. And Small 'i' represents AC quantity.



Problems:

Example 1.1

How much charge is represented by 4,600 electrons?

Solution:

Each electron has -1.602×10^{-19} C. Hence 4,600 electrons will have
 -1.602×10^{-19} C/electron $\times$ 4,600 electrons = -7.369×10^{-16} C

Practice Problem 1.1

Calculate the amount of charge represented by six million protons.

Answer: $+9.612 \times 10^{-13}$ C.

Example 1.2

The total charge entering a terminal is given by $q = 5t \sin 4\pi t$ mC. Calculate the current at $t = 0.5$ s.

Solution:

$$i = \frac{dq}{dt} = \frac{d}{dt}(5t \sin 4\pi t) \text{ mC/s} = (5 \sin 4\pi t + 20\pi t \cos 4\pi t) \text{ mA}$$

At $t = 0.5$,

$$i = 5 \sin 2\pi + 10\pi \cos 2\pi = 0 + 10\pi = 31.42 \text{ mA}$$

Practice Problem 1.2

If in Example 1.2, $q = (10 - 10e^{-2t})$ mC, find the current at $t = 1.0$ s.

Answer: 2.707 mA.

Determine the total charge entering a terminal between $t = 1$ s and $t = 2$ s if the current passing the terminal is $i = (3t^2 - t)$ A.

Example 1.3

Solution:

$$\begin{aligned} Q &= \int_{t=1}^2 i \, dt = \int_1^2 (3t^2 - t) \, dt \\ &= \left(t^3 - \frac{t^2}{2} \right) \Big|_1^2 = (8 - 2) - \left(1 - \frac{1}{2} \right) = 5.5 \text{ C} \end{aligned}$$

The current flowing through an element is

$$i = \begin{cases} 4 \text{ A}, & 0 < t < 1 \\ 4t^2 \text{ A}, & t > 1 \end{cases}$$

Calculate the charge entering the element from $t = 0$ to $t = 2$ s.

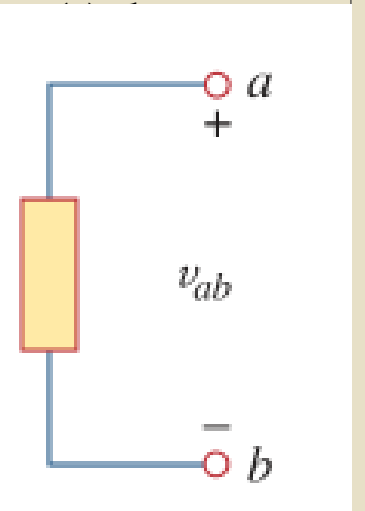
Answer: 13.333 C.

Practice Problem 1.3

Voltage

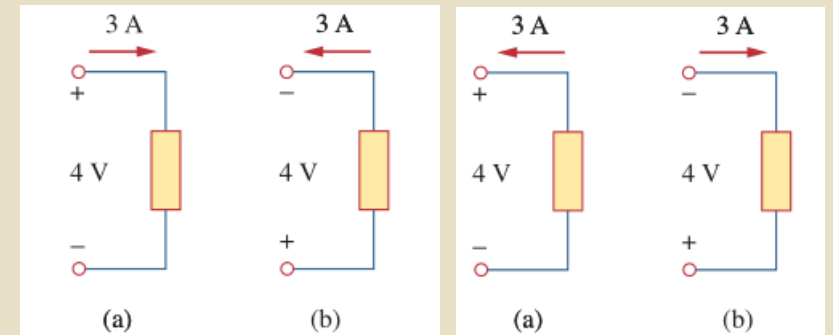
- Voltage (or potential difference) is the energy required to move a unit charge through an element, measured in volts (V).
- to move the electron in a conductor in a particular direction requires some work or energy transfer. This work is performed by an external electromotive force (emf). This emf is also known as voltage or potential difference.
- The voltage v_{ab} between two points a and b in an electric circuit is the energy (or work) needed to move a unit charge from a to b.
- $v_{ab} = \frac{dW}{dq}$; where W is energy in Joule(J), q is charge in Columb (C) and v_{ab} is voltage in volt(V).
- The v_{ab} can be interpreted in two ways: (1) Point a is at a potential of v_{ab} volts higher than point b, potential at point a with respect to point b is v_{ab} . ($v_{ab} = -v_{ba}$)
- v_{ab} = Voltage drop from a to b (a is higher potential and b is lower/ground), and v_{ba} = Voltage rise from b to a.

N:B: Keep in mind that electric current is always through an element and that electric voltage is always across the element or between two points.



Electric Power

- For practical purposes, we need to know how much power an electric device can handle. We all know from experience that a 100-watt bulb gives more light than a 60-watt bulb. We also know that when we pay our bills to the electric utility companies, we are paying for the electric energy consumed over a certain period of time. Thus, power and energy calculations are important in circuit analysis.
- Power is the time rate of expending or absorbing energy, measured in watts (W).
- $p = \frac{dw}{dt} = \frac{dw}{dq} \frac{dq}{dt} = vi$
- If p is positive, power is delivered to a element or absorbed by the element.
- If p is negative, power is supplied by the element. (measured from voltage current polarity)
- Passive sign convention, is satisfied when the current enters through the positive terminal of an element and $p = +vi$. If the current enters through the negative terminal, $p = -vi$.
- the law of conservation of energy, +Power absorbed = - Power supplied



Energy

- the energy absorbed or supplied by an element from time t_0 to time t is $w = \int_{t_0}^t p \, dt = \int_{t_0}^t vi \, dt$
- Energy is the capacity to do work, measured in joules (J).
- The electric power utility companies measure energy in kilo-watt-hours (kWh), 1 kWh= 1 unit.
- 1 kWh=3600 kJ

Problems

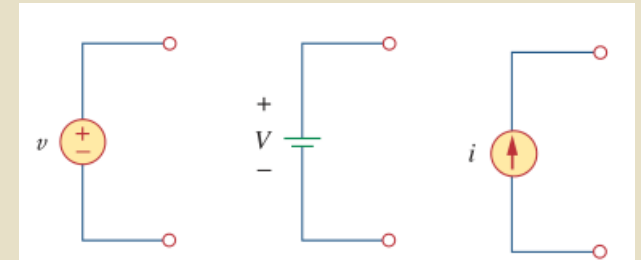
1. An energy source forces a constant current of 2 A for 10 s to flow through a light bulb. If 2.3 kJ is given off in the form of light and heat energy, calculate the voltage drop across the bulb. (Ans: 115 V)
2. To move charge q from point a to point b requires -30 J. Find the voltage drop v_{ab} if: (a) $q = 6\text{C}$ b) -3C . (Ans: a. -5 V, b. 10 V)
3. Find the power delivered to an element at $t = 3\text{ ms}$ if the current entering its positive terminal is $i = 5 \cos 60\pi t$ A and the voltage is: (a) $v = 3i$ (b) $v = 3 \frac{di}{dt}$ (Ans: (a) 53.48 W, (b) -6.396 kW)
4. Find the power delivered to the element in problem 3 at $t = 5\text{ ms}$ if the current remains the same but the voltage is: (a) $v = 2i$ V, (b) $v = (10 + 5 \int_0^t i dt)$ V (Ans: (a) 17.27 W (b) 29.7 W)
5. How much energy does a 100-W electric bulb consume in two hours? (Ans: 200 Wh)
6. A stove element draws 15 A when connected to a 240-V line. How long does it take to consume 180 kJ? (Ans: 50 s)

Circuit Elements

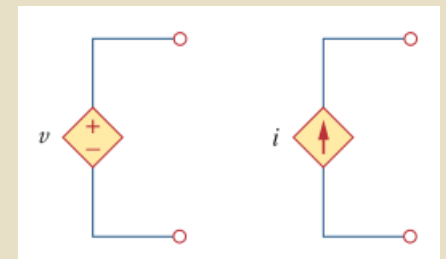
- An electric circuit is simply an interconnection of the elements. Circuit analysis is the process of determining voltages across (or the currents through) the elements of the circuit.
- Two types: a) Passive Elements b) Active elements
- An active element generates and supplies energy, where passive elements absorb energy.
- Passive elements: Resistor, Capacitor, Inductor etc.
- Active elements; Generators, Battery, Op-Amp

Active Elements

- Voltage source and Current source are two most important active elements.
- Based on dependency of others, sources are of two types: a) Independent source b) Dependent Source.
- An ideal independent source is an active element that provides a specified voltage or current that is completely independent of other circuit elements.
- Examples: Battery, Generators etc.
- First symbol of Voltage source represent time varying voltage.



- An ideal dependent (or controlled) source is an active element in which the source quantity is controlled by another voltage or current.
- Denoted by diamond shaped symbol.



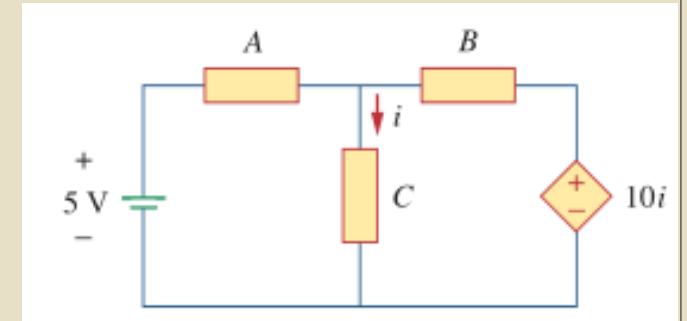
Dependent Source Types:

Four Types:

1. A voltage-controlled voltage source (VCVS).
2. A current-controlled voltage source (CCVS).
3. A voltage-controlled current source (VCCS).
4. A current-controlled current source (CCCS).

Dependent sources are useful in modeling elements such as transistors, operational amplifiers, and integrated circuits.

Students might be surprised that the value of the dependent voltage source is $10i$ V (and not $10i$ A) because it is a voltage source. The key idea to keep in mind is that a voltage source comes with polarities (+/-) in its symbol, while a current source comes with an arrow, irrespective of what it depends on.



Important Notes:

- An ideal voltage source (dependent or independent) will produce any current required to ensure that the terminal voltage is as stated, whereas an ideal current source will produce the necessary voltage to ensure the stated current flow. Thus, an ideal source could in theory supply an infinite amount of energy.
- Not only do sources supply power to a circuit, they can absorb power from a circuit too. For a voltage source, we know the voltage but not the current supplied or drawn by it. By the same token, we know the current supplied by a current source but not the voltage across it.

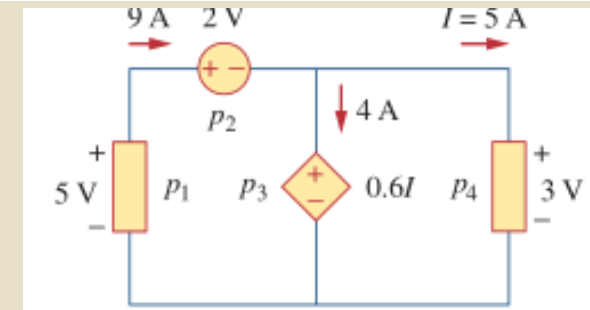
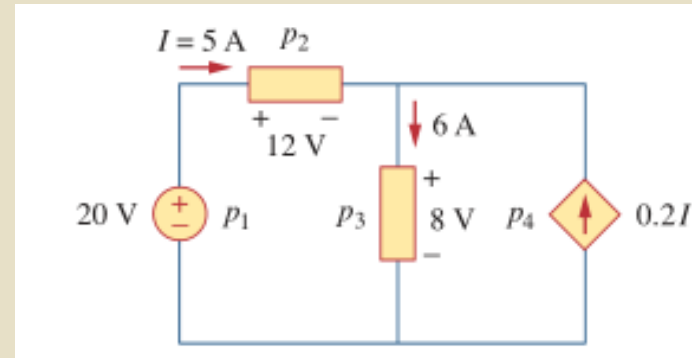
Problem

- Calculate the power supplied or absorbed by each element

Ans: $p_1 = -100$ W (Supplied), $p_2 = 60$ W (Absorbed), $p_3 = 48$ W (Absorbed)

$p_4 = -8$ W (Supplied)

- Compute the power absorbed or supplied by each component of the circuit



Example 1.8

The electron beam in a TV picture tube carries 10^{15} electrons per second. As a design engineer, determine the voltage V_o needed to accelerate the electron beam to achieve 4 W.

Electricity Bill Calc.

A homeowner consumes 700 kWh in January. Determine the electricity bill for the month using the following residential rate schedule:

Referring to the residential rate schedule in Example 1.9, calculate the average cost per kWh if only 350 kWh are consumed in July when the family is on vacation most of the time.

Answer: 14.571 cents/kWh.

Typical average monthly consumption of household appliances.

Appliance	kWh consumed	Appliance	kWh consumed
Water heater	500	Washing machine	120
Freezer	100	Stove	100
Lighting	100	Dryer	80
Dishwasher	35	Microwave oven	25
Electric iron	15	Personal computer	12
TV	10	Radio	8
Toaster	4	Clock	2